the maximum kinetic energy attainable by the moving ball;

the mean current in the external circuit after a long time.

VII. (points) In Fig. , 12 uniform wire rods form a rigid cube of edge . Each rod has resistance . It rotates uniformly about a central axis normal to face at angular speed , in a uniform magnetic field of magnitude normal to that axis. Neglect self-inductance. In Fig. , diagonal makes angle with the field. Find:

the currents in rods , , and ;

the external torque needed to maintain uniform rotation.